MILITARY SPECIFICATION

TEST SET, MISSILE CONTROL SYSTEM AN/AWM-22

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this Specification shall be an integrated group of Contractor Furnished and Government Furnished Special and Standard Support Equipment to facilitate the detailed maintenance of the AN/AWG-10 Missile Control System (MCS) at the Organizational Level.

1.2 Classification - The Test Set, Missile Control System, AN/AWM-22 shall consist of the following items:

Items	Type Designation
Cooler, Air - Hydraulic Unit	HD-747/AWM-22
Test Set, Umbilical Plug	TS-2515/AWM-22
Cart, Maintenance	V-387/AWM-22
Cooler, Air - Rack, Computer	HD-748/AWM-22
Cable Assembly Group	OA-8127/AWM-22
Extension, Slide Rail Maintenance	MX-7345/AWM-22
Test Set, Antenna Alignment	TS-2516/AWM-22

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Items	Type Designation
Filler and Bleeder, Coolant	MX-7346/AWM-22
Test Set, Vertical Gyro - Antenna Alignment	TS-2518/AWM-22
Test Set, Reference Rate Gyro	TS-2512/AWM-21
Test Set, Missile Control System	TS-2511/AWM-21
Generator, Signal	SG-696/AWM-21
Analyzer, Radio Frequency Noise	TS-2517/AWM-22
Fixture, Antenna Transporting	MX-7339/AWM-21
RF Power Test Set	TS-1736/AWM-8
Compressor - Dehydrator, Electronic Equipment	HD-755/AWM-21
Counter, Electronic, Digital	CP-870/AWM-21
Oscilloscope	(to be assigned)
*Multimeter	AN/PSM-4
*Multimeter	TS-505/U

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this Specification to the extent specified herein:

* General and Standard test equipment. This equipment is not supplied as part of the AN/AWM-22 MCS Test Set but is required for use with the AN/AWM-22.

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SPECIFICATIONS

Military

MIL-C-17	Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead
MIL-C-71	Connectors, Coaxial, Radio Frequency, Series N and Associated Fittings
MIL-W-85	Waveguide, Rigid, Rectangular, General Specification for
MIL-C-3432	Cable and Wire, Electrical Power and Control, Flexible and Extra Flexible, 300 and 600 Volts
MIL-C-3767	Connectors, Electrical Power, Bladed Type, General Specification for
MIL-F-3992	Hose and Hose Assembly, Rubber, Air and Vacuum Brake, Automotive
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-H-15362	Harnesses; Test Bench, Avionic Equipment, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for

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MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-T-23336	Technical Directives (Letter Type): Preparation of
MIL-C-24165	Counter, Electronic, 500 Megacycle AN/USM-207
MIL-P-26514	Polyurethane Foam, Rigid or Elastic, for Packaging

STANDARDS

Military

MIL-STD-701	Preferred and Guidance List of Semiconductor Devices
MIL-STD-761	Electric Power, Alternating Current, for Shipboard Use, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-803	Human Engineering Design Criteria for Aerospace Systems and Equipment

DRAWINGS

Bureau of Naval Weapons

58A5A294	Wrench
60A54D116	Gage, Pressure, Dial Indicator
601A54C5	Wrench
677036	Wrench, Pinion

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Military Standard

AN3062	Elbow - Electrical Conduit, 90°
AN3114	Receptacle, External Power, 115/200 volts, 3 Phase
AN3430	Cable and Plug Assembly - External Power, 115/200 volts, 3 Phase
MS3057	Clamp, Cable, Electrical Connector
MS3102	Connectors - Electrical, Receptacles, Box Mounting
MS3106	Connector, Plug, Electric, Straight
MS3108	Connector, Plug, Electric, Angle 90°
MS3120	Connectors, Receptacle, Electric, Crimp-Type, Wall Mounting, Flange, No. 4 Holes, Bayonet Coupling
MS3126	Connectors, Plug, Electric, Crimp Type, Straight, Bayonet Coupling
MS16051	Coupling, Ground Cooling, Combat Type Aircraft
MS17322	Meter, Time Totalizing Miniature, Digital
MS24663	Connector, Plug, Electrical - with Spring Loaded Pivoted Grounding Blade, Polarized
MS25043	Cap, Electrical Connector Receptacle, Dust
MS91236	Connector, Plug, Electrical, UG-21E/U
MS91237	Connector, Receptacle, Electrical, Series N, UG-22E/U and Connector, Plug, Electrical, Series N, UG-23E/U

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Westinghouse Electric Corporation (97942)

305R790	Stud Grounding
575R516	Connector, Plug, Pigmy E type
575R622	Coupling, Quick Disconnect
576R351	Coupling, Quick Disconnect
576R419	Coupling, Quick Disconnect
577R568	Connector, Electrical, Coaxial, Plug, Miniature
577R589	Connector, Receptacle, Electrical, Size 20 Crimp Contacts
577R590	Connector, Receptacle, Electrical, Size 20 Crimp Contacts
577R715	Coupling, Quick Disconnect, Coolant System, 3/8 Tube Size
577R726	Cable Assembly, Special Purpose, Electrical
577R831	Connector, Receptacle, Electrical Polarized, Duplex
577R895	Connector, Plug, Electrical, Insert Half
579R008	Connector, Electrical, Miniature Circular, Special
579R256	Connector, Plug, Electrical, Size 20 Crimp Contact
579R564	Connector, Assembly, Electrical, 25 Coaxial Contacts
579R705	Connector, Receptacle, Electrical

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2.1.1 Availability of Documents - Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office. See 6.4.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and Materials as approved by MIL-T-21200 shall be given first consideration.

(b) Guidelines for the selection of nonstandard parts shall be per MIL-T-21200, Selection of parts and materials.

(c) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass electrical and environmental test.

3.1.1 Nonstandard Parts - Nonstandard parts and materials shall be as outlined in MIL-T-21200, Materials, Parts and Processes, except that mechanical parts shall be considered as nonstandard parts only when the material requirements of MIL-T-21200, as modified herein, are not met.

3.1.2 Tube Types and Semiconductor Types - The choice and application of electron tubes and semiconductor types used in the equipment shall be in accordance with MIL-T-21200 and MIL-STD-701.

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3.1.3 Noncorrosive Hardware - External, Military Standard (MS) hardware which shall be used during the servicing and installation of test equipments shall be of a corrosive-resistant type and shall not be further finish-treated.

3.2 Design and Construction - Contractor Furnished Equipment (CFE) shall conform with all the requirements of MIL-T-21200 and MIL-H-15362, as applicable for design, construction, and workmanship, unless otherwise specified herein. Government Furnished Equipment (GFE) shall conform to the individual equipment specifications stated herein

3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of 24 hours without necessity for readjustment of any operator controls which are inaccessible to the operator during use except where adjustments associated with parts failures are required following replacement of those parts.

3.2.2 Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts.

3.2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200 and MIL-H-15362, as applicable.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in MIL-T-21200.

3.2.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance. The color of front panel indicator lights shall be in accordance with MIL-STD-803.

3.2.7 Calibration - Test and measuring equipment shall be calibrated at periodic intervals established on the basis of stability, purpose and degree of usage. Calibration intervals shall be adjusted as required to assure continued accuracy as evidenced by previous calibration results.

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3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room Ambient: 30° ± 10° Centigrade (C) 86° ± 18° Fahrenheit (F)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input voltage	115 ± 1 Volts (V), 3-phase, 400 cycles per second (cps) ± 2% 27.5 ± 0.5 Volts Direct Current (VDC)

3.2.9 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions, 20 minutes at extreme service conditions.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Function - The AN/AWM-22 MCS Test Set shall provide the necessary equipment to maintain the AN/AWG-10 MCS at the Organizational Level. It shall distribute supplied electrical power and furnish hydraulic power and cooling for operation of the AN/AWG-10 MCS. Special and standard test equipment shall provide the means to perform organization level preventive and unscheduled maintenance and system optimization.

3.3.3 Detail Performance - The Performance of each item of the AN/AWM-22 Test Set can be found per the following table.

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Unit	Paragraph
Cooler, Air - Hydraulic Unit, HD-747/AWM-22	3.4.1
Test Set, Umbilical Plug, TS-2515/AWM-22	3.4.2
Cart, Maintenance, V-387/ AWM-22	3.4.3
Cooler, Air - Rack, Computer, HD-748/AWM-22	3.4.4
Cable Assembly Group, OA-8127/ AWM-22	3.4.5
Extension, Slide Rail Maintenance, MX-7345/AWM-22	3.4.6
Test Set, Antenna Alignment, TS-2516/AWM-22	3.4.7
Filler and Bleeder, Coolant, MX-7346/AWM-22	3.4.8
Test Set, Vertical Gyro - Antenna Alignment, TS-2518/AWM-22	3.4.9
Test Set, Reference Rate Gyro, TS-2512/AWM-21	3.4.10
Test Set, Missile Control System, TS-2511/AWM-21	3.4.11
Generator, Signal, SG-696/AWM-21	3.4.12
Analyzer, Radio Frequency Noise, TS-2517/AWM-22	3.4.13
Fixture, Antenna Transporting, MX-7339/AWM-21	3.4.14
RF Power Test Set, TS-1736/ AWM-8	3.4.15

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Unit	Paragraph
Compressor-Dehydrator, Electronic Equipment, HD-755/AWM-21	3.4.16
Counter, Electronic, Digital, CP-870/AWM-21	3.4.17
Oscilloscope	3.4.18
Multimeter, AN/PSM-4	3.4.19
Multimeter, TS-505/U	3.4.20

3.4 Detail Requirements

3.4.1 Cooler, Air - Hydraulic Unit, HD-747/AWM-22

3.4.1.1 Function - The Cooler, Air - Hydraulic Unit shall accept 3 phase, 400 cps electrical power and distribute power to the normal external connections of the aircraft, and to the Cooler, Air-Rack, Computer (3.4.4), the Compressor-Dehydrator, Electronic Equipment (3.4.16) and to convenience outlets on the Power and Hydraulic Supply Unit (3.4.1.8.2). The Cooler, Air-Hydraulic Unit shall also provide necessary cooling air and hydraulic power for the AN/AWG-10 MCS. The hydraulic power shall be supplied either to the aircraft hydraulic system or to the AN/APG-59 Antenna.

3.4.1.2 Form Factor - The unit shall not exceed 34 inches by 32 inches by 64 inches while in its normal horizontal "in use" attitude. The weight of the unit shall not exceed 425 pounds.

3.4.1.3 Mean Time Between Failures (MTBF) - When maintained in accordance with applicable preventive maintenance procedures, the equipment shall demonstrate at least 3000 hours MTBF. Accept-reject criteria shall be as defined by paragraph 4.2.8.1 of MIL-STD-781.

3.4.1.4 Standard Conditions - An oil temperature of 50° ± 3°C (122° ± 5°F) shall be used to establish normal performance characteristics under the Standard Conditions of 3.2.8 and for making laboratory bench tests.

3.4.1.5 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment, except for the Electronic Equipment Air Cooler (3.4.1.8.4) which will deliver adequate cooling air to the AN/APG-59 Radar Set only over the ambient temperature range of -40° to +30°C (-40° to +86°F).

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3.4.1.6 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-761 Type III. Power requirements, specified below, shall be for the Cooler, Air - Hydraulic Unit only and shall not include the Load Distribution (3.4.1.6.1). Power shall be of the following type and shall not exceed the specified amounts.

(a)	Input Power:	107.5 to 119.5 V, 3-Phase, 380 to 420 cps, 13,500 Volt-Amperes (VA) maximum (max)

3.4.1.6.1 Load Distribution - The equipment shall be capable of distributing power to the following electrical loads:

(a)	Aircraft	-	18,000 VA
(b)	Cooler, Air-Rack, Computer	-	150 VA
(c)	Compressor-Dehydrator	-	500 VA
(d)	Convenience Outlets	-	1,700 VA

3.4.1.7 Noise Level - The noise level of the Cooler, Air - Hydraulic Unit shall not exceed 95 decibels (dB) above reference level of 0.0002 microbar when measured 6 feet from the unit.

3.4.1.8 Contents of Unit - The Cooler, Air - Hydraulic Unit shall consist of the following units:

(a) Dolly Truck

(b) Power and Hydraulic Supply Unit

(c) Hydraulic Hose Assembly

(d) Cooler, Air, Electronic Equipment

3.4.1.8.1 Dolly Truck

3.4.1.8.1.1 Function - The Dolly Truck shall hold and transport the Power and Hydraulic Supply Unit (3.4.1.8.2), the Hydraulic Hose Assembly (3.4.1.8.3) and the Electronic Equipment Air Cooler (3.4.1.8.4).

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3.4.1.8.1.2 Form Factor - The Dolly Truck shall be suitable for one man operation, and shall have two wheels each equipped with 14.7 inch diameter pneumatic tires at one end and two handles at the other end. Rubber feet shall be attached to the frame and shall support the Dolly Truck in a horizontal "in use" attitude. The wheels shall be off the deck in this position. When the Dolly Truck is placed in the "transport" position, the handles shall be elevated a minimum of 27 inches and the forward rubber feet shall be elevated a minimum of 2 inches from the deck. The frame of the Dolly Truck shall provide a wraparound stowage capacity for the Cable Assemblies (3.4.1.8.2.8.1) and the Hydraulic Hose Assembly (3.4.1.8.3). Means shall be provided to keep the Cables and Hoses clear of the wheels and to keep hydraulic hose fittings in place during transport. Four tie-down rings shall be provided on the frame compatible with standard aircraft tie-down devices. The Dolly Truck shall mount the Power and Hydraulic Supply Unit (3.4.1.8.2) on the forward end, with its control panel facing forward. The Power and Hydraulic Supply Unit shall be mounted directly to the frame and shall be removable with the use of small hand tools. The brackets which form attachment points for the Electronic Equipment Air Cooler (3.4.1.8.4) shall be installed on the rear of the Dolly Truck. The Air Cooler shall be mounted on the brackets and shall be removable with the use of small hand tools. The Dolly Truck shall not exceed 34 inches by 21-1/2 inches by 64 inches and its weight shall not exceed 55 pounds.

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3.4.1.8.2 Power and Hydraulic Supply Unit

3.4.1.8.2.1 Function - The Power and Hydraulic Supply Unit shall have a dual hydraulic pressure capability. It shall supply hydraulic power either to the normal aircraft external hydraulic connections or directly to the AN/APG-59 Radar Set hydraulic fittings. In addition the Unit shall provide such electrical power control and distribution as specified herein.

3.4.1.8.2.2 Form Factor - The Power and Hydraulic Supply Unit shall, in its portage condition, be of a rectangular parallelepiped configuration with rounded edges and corners. It shall have a hinged cover over a control panel which shall be accessible to an operator standing in front of the Dolly Truck (3.4.1.8.1.6). Maximum dimensions of the Unit exclusive of hoses, fittings and connectors, shall be 24 inches by 25 inches by 23 inches. Weight, including oil and hoses, shall not exceed 200 pounds.

3.4.1.8.2.3 Performance - The Hydraulic Supply shall be capable of the following:

(a) Pressure - The low pressure supply shall be at least 800 pounds per square inch, gage (psig) but not greater than 1200 psig. The high pressure supply shall be at least 1800 psig but not greater than 3000 psig.

(b) Flow Rate - The flow rate shall be not less than 1.4 gallons per minute (gpm) at 800 psig, and not less than 3.2 gpm at 1800 psig. The flow rate shall be measured at a fluid temperature of 50° ± 3°C (122° ± 5°F).

(c) Hydraulic Fluid - The Unit shall operate with hydraulic fluid per MIL-H-5606.

3.4.1.8.2.4 Contents - The Power and Hydraulic Supply Unit shall consist of the following with indicated characteristics.

(a) Motor Driven Pump - A variable displacement, Motor Driven Pump designed for use with MIL-H-5606 hydraulic fluid shall be provided. The Pump shall be capable of providing pressure and flow rates consistent with applicable Performance (3.4.1.8.2.3) when the motor is energized by power of the type stated in Primary Input Power Requirements (3.4.1.6).

(b) High Pressure Supply Filter - A 5-micron, metal mesh filter of 4 gpm capacity shall be in series with the high pressure supply line. The Filter shall be equipped with a device which

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will cause the Filter (3.4.1.8.2.6(b)) light to illuminate when the high pressure supply is selected and the pressure drop across the filter element reaches 80 ± 10 pounds per square inch (psi).

(c) Low Pressure Supply Filter - A 5-micron, metal mesh Filter of 1.5 gpm capacity shall be in series with the low pressure supply line. The filter shall be equipped with a device which will cause the Filter (3.4.1.8.2.6(b)) light to illuminate when the low pressure supply is selected and the pressure drop across the filter element reaches 80 ± 10 psi.

(d) High Pressure Return Filter - A 105-micron, metal mesh Filter of 4 gpm capacity shall be in series with the high pressure return line.

(e) Pump Drain Filter - A 10-micron Filter of 0.5 gpm capacity shall be provided in the pump drain line.

(f) Purge Tank - A Purge Tank with a minimum capacity of one gallon shall be provided. A sight gage on the right side of the Power and Hydraulic Supply shall be provided to check the fluid level. A Purge Tank Pressure-In hydraulic coupling called out in (3.4.1.8.2.7(a)) shall connect to the Tank. The Tank shall be vented, with an air filter over the vent, and constructed so that spillage shall not occur in any attitude. The fill opening shall have a strainer and sealing cap. In the low pressure system, return oil shall pass through the Purge Tank. In the high pressure system return oil shall bypass the Purge Tank and return directly to the pump. The high pressure system shall be capable of being purged by connecting hoses to the Purge Tank through external connectors.

(g) Heat Exchanger - A Heat Exchanger shall be installed in the low pressure return line before the Purge Tank connection. The Exchanger shall be a liquid-to-air type and limit the fliud temperature to 71°C (160°F) maximum. The Exchanger shall contain a blower powered by a 115 V, 400 cps, 3-phase motor which is capable of providing enough cooling air to meet the fluid temperature requirements against the resistance of the Exchanger plus 0.3 inches of water.

3.4.1.8.2.5 Controls - The following controls shall be provided on the control panel.

(a) MASTER CONTROL - The Master Control switch shall control the primary power to the AN/AWG-10 MCS and, in conjunction with the ACFT ON-ANT ON (3.4.1.8.2.5(b)) and START (3.4.1.8.2.5(c)) controls, shall control power to the Hydraulic Supply.

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(b) ACFT ON - ANT ON (Aircraft or Antenna On) - This control shall be a three pole, double throw switch and shall be interlocked through the MASTER CONTROL (3.4.1.8.2.5(a)) and START (3.4.1.8.2.5(c)) controls. When the switch is placed in the ANT ON position, the ANT ON (3.4.1.8.2.6(c)) lamp shall be illuminated, the low pressure hydraulic supply shall be selected, the PRESS (3.4.1.8.2.6(a)) lamp shall be connected to the low pressure sensing switch, and the FILTER (3.4.1.8.2.6(b)) lamp shall be connected to the Low Pressure Supply Filter (3.4.1.8.2.4(c)) pressure drop sensing device. When the switch is placed in the ACFT ON position, the ACFT ON (3.4.1.8.2.6(d)) lamp shall be illuminated, the high pressure hydraulic supply shall be selected, the PRESS lamp shall be connected to the high pressure sensing switch, and the FILTER lamp shall be connected to the High Pressure Supply Filter (3.4.1.8.2.4(b)) pressure drop sensing device. Selection of the high or low pressure hydraulic supply and illumination of either the ANT ON or ACFT ON indicator lamps shall require prior activation of the MASTER CONTROL and the START controls.

(c) START - The START Control shall be a push button switch which shall be used to start the selected hydraulic supply when the MASTER CONTROL (3.4.1.8.2.5(a)) switch is placed in the activated position.

(d) LAMP TEST - The Lamp Test switch shall be a double pole momentary toggle switch and shall be used to test the continuity of the PRESS (3.4.1.8.2.6(a)) and the FILTER (3.4.1.8.2.6(b)) lamps.

(e) RADAR-RADAR/GEN (Radar-Radar/Computer Rack - The RADAR-RADAR/GEN switch shall be a single pole, double throw toggle switch and shall be used to bypass the Interlock (3.4.1.8.2.8.2.2)circuit around connector (3.4.1.8.2.7(b)) to the Cooler, Air-Rack, Computer (3.4.4) when the Computer Rack is not used. The Interlock circuit shall bypass the connector when the switch is placed in the RADAR position. When the switch is placed in the RADAR/GEN position the Interlock circuit shall operate through the connector.

3.4.1.8.2.6 Panel Indicators - The following indicators shall be provided on the control panel. The colors shall be in accordance with MIL-STD-803.

(a) PRESS (Pressure) - The pressure lamp shall be equipped with a green lens and shall indicate that the hydraulic

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pressure is at least 850 ± 50 psi or 1920 ± 120 psi depending upon the pressure range selected by the ACFT ON-ANT ON (3.4.1.8.2.5(b)) switch. This lamp shall illuminate at its set pressure and remain lighted during operation of the hydraulic supply so long as the selected pressure is maintained.

(b) FILTER - The Filter lamp shall be equipped with red lens, and shall light when the pressure drop across the High Pressure Supply Filter (3.4.1.8.2.4(b)) or the Low Pressure Hydraulic Supply Filter (3.4.1.8.2.4(c)) exceeds 80 ± 10 psi. The lamp shall be connected to the High or Low Pressure Supply Filter by the ACFT ON-ANT ON switch.

(c) ANT ON (Antenna On) - The Antenna On lamp shall be equipped with a green lens and shall light when the Hydraulics (3.4.1.8.2.8.2.1(a)) circuit breaker is closed, the Start (3.4.1.8.2.5(c)) control energized and the low pressure range has been selected by the ACFT ON-ANT ON switch.

(d) ACFT ON (Aircraft On) - The Aircraft On lamp shall be equipped with a green lens and shall light when the Hydraulics circuit breaker is closed, the Start control energized and the low pressure range has been selected by the ACFT ON-ANT ON switch.

(e) COOLING AIR ON - The Cooling Air On lamp shall be equipped with a green lens and shall light when the Cooling Air (3.4.1.8.2.8.2.1(b)) circuit breaker is on.

(f) COOLING AIR OFF - The Cooling Air Off lamp shall be equipped with a red lens and shall light when the Cooling Air circuit breaker is off.

(g) AIRCRAFT POWER ON - The Aircraft Power On lamp shall be equipped with a green lens and shall light when the Master Control (3.4.1.8.2.5(a)) switch is on and the Interlock (3.4.1.8.2.8.2.2) circuit is closed.

(h) AIRCRAFT POWER OFF - The Aircraft Power Off lamp shall be equipped with a red lens and shall light when the aircraft power Master Control switch is off and the Interlock circuit is closed.

(i) COOLING AIR ON TIME - A meter, per MS17322-6A, shall indicate the total time the Electronic Equipment Air Cooler (3.4.1.8.4) is energized.

(j) HYDRAULICS ON TIME - A meter, per MS17322-64, shall indicate the total time the Motor Driven Pump (3.4.1.8.2.4(a)) is energized.

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3.4.1.8.2.7 Connections - The following connectors shall be provided on the Power and Hydraulic Supply Unit:

(a) Hydraulic - There shall be five hydraulic couplings as follows: 576R351H08 for the antenna low pressure supply, 576R351H18 for the antenna low pressure return, 575R622H02 for the aircraft high pressure supply, 576R419H02 for the aircraft high pressure return, and 576R351H08 for Purge Tank Pressure-In.

(b) Electrical - A primary power input receptacle, a recessed AN3114-1B male connector, shall be provided. One connector, 577R589H03, shall be provided for the Cooler, Air-Rack, Computer (3.4.4) interlock and power circuit. Two duplex, grounded, convenience power outlets, 577R831H01, shall be provided to distribute primary power to auxiliary test equipment. A connector, MS3102-R24-6S, shall be provided for the Electronic Equipment Air Cooler (3.4.1.8.4). A grounding stud, 305R790H01, shall be provided. One receptacle, UR107F, per MIL-C-3767/9, shall be provided for the Compressor-Dehydrator, Electronics Equipment, (3.4.16).

3.4.1.8.2.8 Electrical Components

3.4.1.8.2.8.1 Cable Assemblies - Two cable assemblies shall be provided as follows:

(a) Power Cable Assembly - The Power Cable Assembly to the aircraft shall be similar to AN3430 except that it shall have four No. 4 AWG wires for phase A, B, C and neutral of the 3-phase, 400 cps, 115/200 V power and two No. 12 AWG wires for control purposes. One end of this Cable Assembly shall be permanently attached to the Power and Hydraulic Supply Unit through an AN3062-32 elbow and MS3057-32A clamp, which shall allow coiling the cable about the unit. The length of this cable shall be 216 ± 6 inches.

(b) Grounding Cable Assembly - The Grounding Cable Assembly shall consist of a 96 ± 6 inch length of No. 4B and S gage flexible cable, one end of which shall be terminated with a spade lug compatible with the grounding stud of 3.4.1.8.2.7(b), and the other end shall be terminated with a 200-ampere (A) battery clip.

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3.4.1.8.2.8.2 Protective Devices

3.4.1.8.2.8.2.1 Circuit Breakers - Five Circuit Breakers shall be provided as follows:

(a) HYDRAULICS - The Hydraulics circuit breaker shall be of a mechanically-coupled, 6 pole, manual reset type. Three poles shall control power to the Heat Exchanger (3.4.1.8.2.4(g)) motor and shall have a maximum continuous rating of 1.5 amps. These three poles shall be connected to the output side of a second set of three poles which shall also control power to the Motor Driven Pump (3.4.1.8.2.4(a)). These three poles shall have a maximum continuous rating of 35 A. The circuit breaker shall be suitable for 115/200 V, 3-phase, 400 cps operations.

(b) COOLING AIR - The Cooling Air circuit breaker shall control power to the Electronic Equipment Air Cooler (3.4.1.8.4) and shall be a 3 pole, manual reset type having a continuous rating of 15 A. The circuit breaker shall be suitable for 115/200 V, 3 phase, 400 cps operation and shall be used to turn the Electronic Equipment Air Cooler on and off. The circuit breaker shall have a set of auxiliary contacts which shall be used to energize the Cooling Air OFF (3.4.1.8.2.6(f)) light.

(c) DATA GEN POWER (Computer Air-Rack Power) - The DATA GEN POWER circuit breaker shall control power to the Cooler, Air-Rack, Computer (3.4.4) and shall be a 3 pole, manual reset type having a continuous rating of 1.5 A. The circuit breaker shall be suitable for 115/200 V, 3 phase, 400 cps operation.

(d) UTILITY POWER - The Utility Power circuit breaker shall control power to the Convenience Power Outlets of 3.4.1.8.2.7(b) and shall be a single pole, manual reset type with a continuous rating of 15 A. The circuit breaker shall be suitable for 115 V, single-phase, 400 cps operation.

(e) COMPRESSOR-DEHYDRATOR - The Compressor-Dehydrator circuit breaker shall control power to the Electronic Equipment Compressor-Dehydrator (3.4.16) and shall be a 3 pole, manual reset type having a continuous rating of 1.5 A. The circuit breaker shall be suitable for 115/200 V, 3-phase, 400 cps operation.

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3.4.1.8.2.8.2.2 Interlock - An Interlock circuit from the Cooler, Air-Rack, Computer (3.4.4) which senses the loss of Computer cooling air, shall be routed from the Cooler, Air-Rack, Computer through the Power and Hydraulic Supply Unit to the aircraft. The loss of Computer cooling air shall interrupt the Interlock circuit and remove power to the Aircraft. The Interlock is operative only when the Cooler, Air-Rack, Computer is connected to the Power and Hydraulic Supply Unit and the RADAR-RADAR/GEN 3.4.1.8.2.5(e) switch is in the RADAR/GEN position.

3.4.1.8.3 Hydraulic Hose Assembly

3.4.1.8.3.1 Function - The Hydraulic Hose Assembly shall deliver hydraulic power from the Power and Hydraulic Supply Unit (3.4.1.8.2) to either the aircraft or the AN/APG-59 Antenna.

3.4.1.8.3.2 Form Factor - The Hydraulic Hose Assembly shall consist of two hoses, each 216 ± 6 inches in length, less adapters and couplings. The two individual hoses shall be formed into a harness by enclosing with heat shrinkable, neoprene tubing for a length of 167-1/2, + 1 inches. Both ends of each hose shall be free. The hoses shall be additionally wrapped with vinyl tape and banded with stainless steel clamps at each end of the joined portion and at a nominal 8 inches from each end of the joined portion. Hanging chains shall be attached to the two end bands. The Supply Hose shall be teflon-lined with a carbon steel, inner wire braid and a wire braid outer cover and shall be terminated with a 575R622H03 coupling at one end and a 576R351H07 coupling at the other end. The Return Hose shall be teflon-lined with a single-layer wire braid and shall be terminated with a 576R419H03 coupling at one end and a 576R351H17 coupling at the other end. Fittings shall be of the self-sealing, quick disconnect type. The weight of the Hydraulic Hose Assembly shall not exceed 60 pounds.

3.4.1.8.4 Cooler, Air, Electronic Equipment (Air Cooler)

3.4.1.8.4.1 Function - The Air Cooler shall supply cooling air to the AN/AWG-10 MCS during organizational level maintenance through the normal ground cooling connection of the aircraft.

3.4.1.8.4.2 Form Factor - The Air Cooler shall be of generally cylindrical configuration with sub-units concentrically assembled along the cylinder axis. Maximum dimensions of the unit shall be 18 inches in diameter and 25 inches long, less the Flexible Air Duct (3.4.1.8.4.4(b)) which shall be 150 ± 6 inches long including coupling. The weight of the Air Cooler including the Flexible Air Duct shall not exceed 55 lbs.

3.4.1.8.4.3 Performance - The air flow shall be a minimum of 380 cubic feet per minute into a pressure of 25 inches of water.

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3.4.1.8.4.4 Contents - The Air Cooler shall consist of the following with indicated characteristics.

(a) Cooler Unit - The Cooler Unit shall consist of a Base, an Intermediate Housing and an Output Air Housing packaged as a unit. The Base shall be a cylindrical compartment and shall be lined with sound absorbent material. It shall provide a removable dust filter fitted with a screen guard for mechanical protection and shall mount the cylindrical Intermediate Housing. The Intermediate Housing shall be equipped with an 115/200 V, 400 cps, 3-phase, electrical motor to drive a vane axial fan compatible with Performance (3.4.1.8.4.3) and Primary Input Power Requirements (3.4.1.6). An anti-shock wave cone shall also be furnished in the air output chamber. The Intermediate Housing shall provide a carrying handle, and electrical receptacle (MS3102E24-6P), a combination overload circuit breaker and off/on switch, and a removable, rain/splash filter shield. The Output Air Housing shall be of metallic construction and shall be compatible with the Intermediate Housing Assembly and the Flexible Air Duct (3.4.1.8.4.4(b)).

(b) Flexible Air Duct - The Flexible Air Duct shall be 150 ± 6 inches long including coupling, and shall be 5 inches, nominal, in diameter. One end shall have a semi-rigid cuff which shall mate with the output air facility of the Cooler Unit (3.4.1.8.4.4(a)). This cuff shall incorporate a two piece, quick-detechable clamp. The other end of the Duct shall have a transition piece from a 5-inch to 4-inch diameter and shall terminate in an MS16051-1 coupling. The weight of this item shall not exceed 20 pounds.

(c) Cable - The primary input cable shall be 72 ± 6 inches long. One end shall have a connector (MS3108R24-6S) compatible with the Cooler Unit (3.4.1.8.4.4(a)) connector and the other end shall have a connector (MS3106R24-6P) compatible with the Air Cooler connector on the Power and Hydraulic Supply Unit (3.4.1.8.2). The cable shall be a 7 conductor, flexible cable per MIL-C-3432. Connectors shall be wired pin to pin except pin H which shall be left blank.

3.4.2 Test Set, Umbilical Plug, TS2515/AWM-22

3.4.2.1 Function - The equipment shall provide the means, in conjunction with the Selector-Test Programmer, to verify the presence and quality

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of the signals at aircraft umbilical connectors, the firing circuits and three Klystron Tuning Circuits of the AN/AWG-10 MCS for the following type missiles:

(a) AN/AIM-7D

(b) AN/AIM-7E

(c) AN/AIM-9B

(d) AN/AIM-9D

3.4.2.2 Form Factor - The Umbilical Plug Test Set shall be a maximum of 27 inches by 17 inches by 15 inches. The weight of the Test Set, including the transit case, shall not exceed 100 pounds.

3.4.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.2.4 Primary Input Power Requirements

3.4.2.4.1 Voltage Limitation - The equipment shall meet all applicable requirements of MIL-T-21200 for Class 2 equipment, except that the power required shall be of the following types and shall not exceed the specified amounts:

(a) 107.5 to 119.5 V, Single Phase, 380 to 420 cps, 5.0 VA

(b) 24 to 29 VDC, 300 watts.

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3.4.2.4.2 Performance Limitation - The equipment shall also operate without damage, but not necessarily with specified performance with the following input voltages:

(a) 103.5 to 107.5 V and 119.5 to 126.5 V, single-phase 380 to 420 cps

3.4.2.5 Contents of Unit - The Umbilical Plug Test Set shall consist of the following items:

	Items	Quantity
(a)	Umbilical Test Plug (Sparrow)	4
(b)	Umbilical Test Plug (Sparrow and Sidewinder)	4
(c)	Aero 7A Launcher Foot Block	4
(d)	Gage, Pressure, Dial Indicator	1
(e)	Carrying Basket	2
(f)	3/8 inch Hexagonal Wrench	1
(g)	5/16 inch Hexagonal Wrench	1
(h)	Missile Gone Switch Cap	4
(i)	Cable	4
(j)	Transit Case	1

3.4.2.5.1 Umbilical Test Plug (Sparrow and Sparrow/Sidewinder)

3.4.2.5.1.1 Function - The Umbilical Test Plug shall, when connected to the aircraft umbilical and upon command from the Selector-Test Programmer, provide the means to sequentially connect the various wires of the aircraft umbilical to the Selector-Test Programmer to ascertain that the aircraft umbilical and firing circuit relays are operating properly. Four of the Umbilical Test Plugs shall function with the Sparrow missile launch system only (Aero 7A Launcher or LAU-17/A Missile Pylon). Four shall function with both the Sparrow and the Sidewinder missile launch system (including the LAU-7/A Launcher) with Power Supply PP 2315A. The Umbilical Test Plug shall generate, switch, transmit, and terminate the appropriate voltages to

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ascertain, in conjunction with the Selector-Test Programmer, the proper functioning of the Firing Circuits, and Klystron Tuning Signals. An Umbilical Test Plug must be connected to every umbilical connector on the aircraft before the tests can be performed.

3.4.2.5.1.2 Form Factor - Each Umbilical Test Plug shall be 12 inches by 5 inches by 5 inches maximum. The weight of each Umbilical Test Plug shall not exceed 6.0 pounds. Each shall be yellow in color, with red stripes. The following notice shall be prominently displayed on the side of the Sparrow/Sidewinder Umbilical Test Plug:

CAUTION

Before connecting 59J2, install Insulating Cover on 59J1.

3.4.2.5.1.3 Inputs - The following inputs shall be required for Umbilical Test Plug operation. These inputs shall be exclusive of the signals to be monitored.

(a) Homing Command - A +28 VDC return shall be applied to 59J1* or +28 VDC shall be applied to 59J2* to establish the initial position of the Umbilical Test Plug signal selector at the initiation of the test. 59J1 and 59J2 are defined in 3.4.2.5.1.12.

(b) DC Power Source - +28 VDC shall be applied continuously to 59J1 or 59J2.

(c) Stepping Command - A +28 V pulse of 100 milliseconds (ms) duration or greater shall be applied to 59J1 or 59J2 through a 7.5 kilohms (K) resistor to advance the stepping switch, 59S2, one step.

3.4.2.5.1.4 Outputs - The Umbilical Test Plug shall provide the following output signals.

* Note - References to connector 59J1 throughout 3.4.2.5.1 shall apply to both the Umbilical Test Plug (Sparrow) (3.4.2.5(a)) and the Umbilical Test Plug (Sparrow and Sidewinder) (3.4.2.5(b)) when they are used with the Sparrow Launch System. References to connector 59J2 throughout 3.4.2.5.1 shall apply only to the Umbilical Test Plug (Sparrow and Sidewinder) (3.4.2.5 (b)) when used with the Sidewinder Launch system.

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(a) Interlock - To prevent accidental missile firing in the event a missile is inadvertently left aboard, interlocks shall be provided by jumpering the appropriate pins on 59J1 with a 15 amp fuse and jumpering the appropriate pins on 59J2 with hookup wire.

(b) Selected Test Signal - The particular test signal selected by the Umbilical Test Plug Signal Selector shall appear on 59J1 or 59J2. The signals shall be as specified in 3.4.2.5.1.5.

(c) AN/AIM-9D/9B EPU Signals - A 6.0 ± 0.5 kilocycles per second (kc), 22.5 ± 2.5 Volts root-mean-square (VRMS) output shall be provided at 59J2. This signal shall simulate the Electronic Power Unit (EPU) activate signal from the AN/AIM-9B or AN/AIM-9D missile.

(d) AN/AIM-7D/7E Trigger Signal - The AN/AIM-7D/7E Trigger Signal on 59J1 shall be jumpered to the EPU Squib signal on 59J1. This shall permit the arrival of the EPU signal at the aircraft umbilical connector to serve as the return of Trigger Signal to the missile firing circuits.

(e) AN/AIM-7D/7E Phase A Power - Upon loss of phase A power on 59J1, +28 VDC shall appear on 59J1 via the signal selector.

(f) AN/AIM-7D/7E EPU Signal - Upon loss of +28 VDC on 59J1, +28 VDC shall appear on 59J1 via the signal selector.

(g) Klystron Tuning Drive - The following signals shall be provided on 59J1 to test the Tuning Drives TG-75/APA-128 and TG-76/APA-128.

(1) Recycle signals per 3.4.2.5.1.10(a)

(2) AFC signals per 3.4.2.5.1.10(b)

(3) Limit signals per 3.4.2.5.1.10(c)

(h) Sidewinder Aboard Signal - A ground shall be placed on 59J2 which shall simulate a Sidewinder missile aboard.

3.4.2.5.1.5 Signals Selected - The Umbilical Test Plug signal selector shall choose the following signals and present them to the Selector-Test Programmer. Signals shall be made to appear on connector 59J1 or 59J2 dependent upon the type missile tested.

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3.4.2.5.1.5.1 AN/AIM-7D/7E Signals - The following signals shall be made to appear on connector 59J1 of the Sparrow or Sparrow and Sidewinder Umbilical Test Plugs.

(a) Head Aim Pitch

(b) Head Aim Yaw

(c) English Bias Pitch

(d) English Bias Yaw

(e) Altitude 1

(f) Altitude 2

(g) AN/AIM-7E (Switch after Boost)

(h) AFC

(i) EPU Squib and Hydraulic Igniter

(j) Stepping Ground

(k) System Ground

(l) Fuse Out

(m) Launch Signal

(n) Simulated Doppler

(o) Roll Command

(p) Sweep Select

(q) Sweep Control

(r) Klystron Tune

(s) Initial English Bias Pitch

(t) Initial English Bias Yaw

(u) 115 V, 400 cps, Phase B

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(v) 115 V, 400 cps, Phase C

(w) Umbilical Retract

(x) Recycle

(y) Klystron Limit

(z) Klystron (Test) Interlock

3.4.2.5.1.5.2 AN/AIM-9B/9D Signals - The following signals shall be made to appear on connector 59J2 of the Sparrow and Sidewinder Umbilical Test Plug only.

(a) Missile B+

(b) Disable B+

(c) Missile Filament

(d) 134 VRMS, 6 kc

(e) Ground

(f) Ground

(g) Gas Grain Squib

(h) Stepping Ground

(i) Power Change Over

(j) Servo Heater

(k) +28 VDC N_2 Valve

(l) Launch Latch

(m) 30 V, 400 cps, Phase 3

(n) 30 V, 400 cps, Phase 2

(o) 30 V, 400 cps, Phase 1

(p) AGC Disabling

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(q) Caging Voltage Return

(r) 115 V, 400 cps

(s) Caging Voltage

(t) Power Interlock

3.4.2.5.1.5.3 Cartridge Signals - The ejection cartridge signals in the Aero 7A Launcher shall be made to appear on connectors 59J5 and 59J6.

3.4.2.5.1.5.4 AN/AIM-7D/7E Motor Fire - A signal shall indicate completion of the Sparrow Motor Fire Test (3.4.2.5.1.6) and shall be provided to the Selector-Test Programmer.

3.4.2.5.1.5.5 AN/AIM-9B/9D Motor Fire - A signal shall indicate completion of the Sidewinder Motor Fire Test (3.4.2.5.1.7) and shall be provided to the Selector-Test Programmer.

3.4.2.5.1.5.6 Power Detect Output - The output of the AN/AIM-7D/7E and AN/AIM-9B/9D Power Detector (3.4.2.5.1.8 and 3.4.2.5.1.9) shall be provided to the Selector-Test Programmer.

3.4.2.5.1.6 AN/AIM-7D/7E Motor Fire Load Test - The Umbilical Test Plug shall provide the circuit to draw 9.0 ± 1.5 amperes through the AN/AIM-7D/7E Motor Fire Circuit Breaker (3.4.2.5.1.11(d)) when energized by a 28 VDC Motor Fire Signal. The Circuit Breaker shall be a delay type to trip after 15 milliseconds. A signal proportional to the Motor Fire Signal shall also be switched directly to the Selector-Test Programmer.

3.4.2.5.1.7 AN/AIM-9B/9D Motor Fire Load Test - The Umbilical Test Plug shall provide a circuit to draw 3.45 ± 0.4 A when energized by a 28 VDC AIM-9B/9D Motor Fire Signal having a 5-ohm source impedance. A signal proportional to the Motor Fire Signal shall also be switched directly to the Selector-Test Programmer.

3.4.2.5.1.8 AN/AIM-7D/7E Power Detector - To detect the absence or loss of AC power, the Umbilical Test Plug shall provide the following voltages at 59J1:

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(a) -9.1 ±1 VDC when excited by 115 V, 400 cps, phase A at 59J1.

(b) -10.0 ±1 VDC when excited by 115 V, 400 cps, phase B at 59J1.

(c) -10.0 ±1 VDC when excited by 115 V, 400 cps, phase C at 59J1.

3.4.2.5.1.9 AN/AIM-9B/9D Power Detector - To detect the absence or loss of AC power, the Umbilical Test Plug shall provide the following voltages at 59J2:

(a) -10 ± 1 VDC when excited by 115 V, 400 cps appearing 59J2.

(b) -2.6 ±0.3 VDC when excited by 30 V, 400 cps, phase 2 at 59J2.

(c) -2.6 ±0.3 VDC when excited by 30 V, 400 cps, phase 3 at 59J2.

(d) -5.0 ±1 VDC when excited by 30 V, 400 cps, phase 1 at 59J2.

3.4.2.5.1.10 Umbilical Test Plug Simulations - The following Klystron Tuning Drive test voltages shall be generated and transmitted to the system:

(a) Recycle

(1) +5.1 ±0.5 VDC

(2) +3.81 ±0.09 VDC

(3) +2.69 ±0.09 VDC

(4) 0.0 ±0.005 VDC

Source Impedance of the voltage to be simulated shall be 150 K. The Simulated Doppler (-6 VDC) signal on 59J1 shall be switched to the Selector-Test Programmer on the +3.81, +2.69 and 0.0 VDC recycle simulations.

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(b) AFC (Automatic Frequency Control)

(1) +2.21 ±0.09 VDC

(2) +0.975 ±0.075 VDC

(3) -2.21 ±0.09 VDC

(4) -0.975 ±0.075 VDC

Source Impedance of the voltage to be simulated shall be 75 K. The 5.1 ±0.5 VDC Recycle (3.4.2.5.1.10(a)(1)) voltage shall be applied to the Recycle input simultaneously with the four AFC voltages. The Tuning Drive Motor Signal (59J1) shall be switched to the Selector-Test Programmer during this time.

(c) Limit

(1) +5.1 ±0.5 VDC

(2) -5.1 ±0.5 VDC

Source Impedance of the voltage to be simulated shall be 12 K. The 0 ±0.005 VDC Recycle (3.4.2.5.1.10(a)(4)) voltage shall be applied simultaneously with the two Limit voltages. The Tuning Drive Motor Signal (59J1) shall be switched to the Selector-Test Programmer during this time.

3.4.2.5.1.11 Controls

(a) AN/AIM-7D/7E Motor Fire Circuit Breaker - A Circuit Breaker shall be provided with Off-Reset position markings to perform the Motor Fire Load Test specified in 3.4.2.5.1.6.

3.4.2.5.1.12 Connectors - The following connections to the Umbilical Test Plugs shall be provided:

	Function	Pigtail Length (Inches)	Reference Designation	Receptacle Type
(a)	AN/AIM-7D/7E Inputs	-	59J1	577R895H01
(b)	AN/AIM-9B/9D Inputs	-	59J2	577R589H36

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	Function	Pigtail Length (Inches)	Reference Designation	Receptacle Type
(c)	AN/AIM-7D/7E Motor Fire	24 ±6	59J3 59J4	577R726H01
(d)	Jettison Firing Input (Wing)	24 ±6	59J5	575R516H03
(e)	Jettison Firing Input (Fuselage)	24 ±6	59J6	575R516H13
(f)	AN/AIM-9B/9D Striker Points	-	59J7	-

3.4.2.5.2 Aero 7A Launcher Foot Block - Four Launcher Foot Blocks shall be provided to block down the Aero 7A Launcher Foot 8 ± 1/4 inches to close the reel operated switch. The Foot Block shall be a 9-inch long, 160-degree section of a 1-5/8 inch outside diameter by 1/8 inch thick aluminum cylinder. The weight of this Block shall not exceed 1/2 pound.

3.4.2.5.3 Gage, Pressure, Dial Indicator - One Gage, Pressure, Dial Indicator, MX-3298/ASM-20 shall be provided to perform a static pressure check of the gas system in the Sidewinder LAU-7A Launcher.

3.4.2.5.4 Carrying Baskets - Two compartmented, carrying baskets, each to hold four Umbilical Test Plugs (3.4.2.5.1), one Wrench (3.4.2.5.5) or (3.4.2.5.6), two AIM-9B/9D Cables (3.4.2.5.8), two Missile Gone Switch Caps (3.4.2.5.7), and two Aero 7A Launcher Foot Blocks (3.4.2.5.2) shall be provided. The Carrying Basket shall be provided with a carrying handle, shall be of welded wire and sheet metal construction, and shall not exceed 21 inches by 7 inches by 9 inches. The weight of each Basket shall not exceed 9 pounds.

3.4.2.5.5 Wrench - One 3/8 inch, hexagonal Wrench shall be provided in each kit.

3.4.2.5.6 Wrench - One 5/16 inch hexagonal Wrench shall be provided in each kit.

3.4.2.5.7 Missile Gone Switch Cap - Four threaded Caps, per MS25043-10, shall be provided to close the Missile Gone switches of the Aero 7A Missile Launcher.

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3.4.2.5.8 Cables - Four AIM-9B/9D Cables shall be Supplied with each kit. The Cables shall be 48 ± 6 inches in lenght, and shall terminate connector 59J2 and AIM-9B/9D Umbilical Input connector. One end shall have a 577R590H31 connector and the other end shall have a 579R008H01 connector.

3.4.2.5.9 Transit Case - A Transit Case shall be provided to hold the two Carrying Baskets (3.4.2.5.4). The maximum dimensions of the Transit Case shall be 27 inches by 17 inches by 15 inches. The weight of the Transit Case shall not exceed 30 pounds. Cushioning shall be provided in the Transit Case and shall be polyurethane foam, charcoal gray color, and 2 pound density per MIL-P-26514 Type I.

3.4.3 Cart, Maintenance, V387/AWM-22

3.4.3.1 Function - The equipment covered by this specification shall be for Organizational and Intermediate Level use and shall be used to transport test equipment and maintenance kits to the aircraft and to provide steps for high reach when servicing the AN/AWG-10 MCS.

3.4.3.2 Form Factor - The Maintenance Cart shall be a two wheeled, hand truck device, providing a ladder, non-skid supports, shelves for test equipment, tie-down rings and cable storage. The structure shall be formed so that when the unit is in its stowed position, it will rest on non-skid pads and the wheels shall be off the deck. Tie-down rings shall be provided to secure the Maintenance Cart to the deck. The wheels shall be equipped with 14.7 inch pneumatic tires. The compliance of the tires shall permit the loaded Cart to pass over a 1-1/2 inch diameter cable without excessive operator force. The center of gravity of the loaded Cart shall be located so that the Cart will not pitch over if the wheels strike an obstacle. Non-skid type pads shall be provided at the deck contact points for the stow position and the standing position. The ladder shall have three steps, 20 inches long and 3 inches wide, with a nominal 10 inch riser. Pivots shall allow the ladder to retract and stow beneath the Cart. The Cart shall be capable of being erected so that the wheels clear the deck, and the rear load is supported by the ladder. In this position the test equipment mounted on the Cart shall be conveniently located for use. Three shelves of 20 inches minimum width shall be provided to mount test equipment. The shelves shall be mounted between the truck rails and located so that equipment on each shelf will face the operator when the truck is standing with the ladder in position. The equipment shall be secured to the shelf it is located on. The shelves shall not contribute to the structural strangeth of the cart. Each shelf shall be capable of supporting a load of 100 pounds, evenly distributed. The Cart shall have a minimum load carrying capacity of 200 pounds. The cart shall provide a storage

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location for securing the Slide Rail Maintenance Extension (3.4.6). A small bin shall be included to contain any test leads and service manuals that may be necessary for use with the equipment on the Cart. The maximum outline dimensions of the Cart in the stowed position shall not exceed 30 inches by 36 inches by 65 inches. The total weight of the Maintenance Cart, including cables, shall not exceed 200 pounds.

3.4.3.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combinations of these conditions as specified in MIL-H-15362.

3.4.3.4 Power Connections - Four standard single phase, 115 V, 400 cps, Convenience Power Outlets shall be provided to connect test equipment mounted on the Maintenance Cart. A 3-wire power cable, 25 ± 1 feet in length, shall be provided. The cable shall terminate in a 3-prong, male plug, MS24663-1. The cable and outlets shall have a current carrying capacity of 15 A.

3.4.4 Cooler, Air-Rack, Computer, HD-748/AWM-22 (Computer Rack)

3.4.4.1 Function - The Computer Rack shall provide mounting and cooling for the Velocity, Launch, and Pilot Command Signal Computers of the AN/APG-59 Radar Set and shall provide for operation of these units when they are removed from the aircraft.

3.4.4.2 Form Factor - The Computer Rack shall consist of a vertical plenum chamber mounted on a four legged base. Each AN/APG-59 Computer shall mount to one of three sides of the plenum by its normal mounting points. Three removable Plenum Covers (3.4.4.6(e)) shall be furnished, each to cover one set of Computer air ports, so that there shall either be a Computer or a Cover mounted on each of the three sides of the Computer Rack. The maximum dimensions of the Computer Rack shall be 17 inches by 25 inches by 31 inches. Total weight shall not exceed 40 pounds.

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3.4.4.3 Performance - The Computer Rack shall deliver to the Computers a minimum of 1.25 pounds per minute of air at a pressure of 5 inches of water.

3.4.4.4 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-H-15362.

3.4.4.5 Primary Input Power Requirements - The equipment power requirements shall be: 103.5 to 126.5 V, three-phase, 380 to 420 cps, 50 VA max per phase.

3.4.4.6 Contents of Unit - The Air Cooler Computer Rack shall consist of the following with indicated characteristics.

(a) Air Filter - The Air Filter shall be a readily cleaned, oil treated, permanent type. The Filter shall have a ventilation area of 70 square inches and a size of 1/2 inch by 8 inches by 11 inches.

(b) Blower - The Blower shall be a vaneaxial fan with an integral, 3-phase, 4 wire, 115 V line-to-neutral, 400 cps motor. The Blower shall deliver air into the plenum per Performance (3.4.4.3). Maximum cylindrical envelope dimensions shall not exceed 3-1/2 inches long by 4 inches in diameter. Weight shall not exceed 2 pounds.

(c) Connector - The electrical connection to the Computer Rack shall be made through a 577R589H01 connector.

(d) Air Interlock - An air pressure sensing switch shall be installed. The switch shall be set to open when the air pressure within the plenum chamber decreases to a pressure of 5.5 -0.00, +0.25 inches of water. The output of the sensing circuit shall operate as a control interlock through the Cooler Air-Hydraulic Unit (3.4.1) for removal of power to the aircraft in the event of loss of cooling air supplied to the Computer units mounted on the Computer Rack.

(e) Plenum Covers - Three removable Plenum Covers, one for each Computer unit, shall be provided for the three plenum surfaces. These Covers shall block the Computer cooling air ports in the absence of a Computer. The Cover shall

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perform the same function on the Computer air plenum in the aircraft. The nominal dimensions of two of the Covers shall be 10 inches by 7 inches by 1-3/16 inches. The nominal dimensions of the third Cover shall be 19 inches by 5-3/8 inches by 1-3/16 inches. The weight of each Cover shall not exceed 3.0 pounds.

(f) Cable Assembly - A Cable Assembly, 30 ± 1 feet long, shall be provided. The Cable Assembly shall be terminated at one end in a 577R590H03 connector and a 577R590H02 connector at the other end.

3.4.4.7 Associated Equipment - The Computer Rack is intended to work in conjunction with the Cooler, Air - Hydraulic Unit (3.4.1). The Cooler, Air - Hydraulic Unit distributes electrical power to the Computer Rack and interrupts the primary power to the AN/APG-59 Radar Set in the event the Air Interlock (3.4.4.6(d)) circuit shall open and is not bypassed.

3.4.5 Cable Assembly Group, OA-8127/AWM-22

3.4.5.1 Function - The Cable Assembly Group shall provide the patch cables necessary to operate the specified components of the AN/APG-59 Radar Set as part of the system when they are removed for servicing from their normal mounting position in the aircraft.

3.4.5.2 Form Factor - The Cable Assembly Group shall be 40 inches by 30 inches by 21 inches maximum. The total weight of the Cable Assembly Group shall not exceed 230 pounds.

3.4.5.3 Cabling and Connections - Cables, cabling, and grounding of cable shields shall be compatible with the requirements of the AN/APG-59 Radar Set.

3.4.5.4 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-H-15362.

3.4.5.5 Contents of Unit - The Cable Assembly Group shall consist of one each of the following:

(a) Command Radar Indicator Cable Assembly - This Cable Assembly shall permit normal operation within the system of the Command Indicator Unit or Radar Indicator Unit, when either one is removed from the aircraft. The Cable Assembly shall consist of two individual cables 18 feet ± 7 inches in length. The individual cables shall be formed into one main assembly leaving the individual cables free at both ends for a

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length of 18 ± 1 inches. One cable shall have a 577R590H45 connector on one end and a 579R256H41 connector on the other end to mate with system connectors 9P1 or 10P1 and 9J1 or 10J1 respectively. The other cable shall have a 577R590H36 connector on one end and a 579R256H31 connector on the other end to mate with system connectors 9P2 or 10P2 and 9J2 or 10J2 respectively.

(b) Selector-Test Programmer Cable Assembly - This Cable Assembly shall permit normal operation within the system of the Selector-Test Programmer when it is removed from the aircraft. The Cable Assembly shall consist of two individual cables 18 feet ± 7 inches in length. The individual cables shall be formed into one main assembly leaving the individual cables free at both ends for a length of 18 ± 1 inches. One cable shall have a 577R590H64 connector on one end and a 579R256H59 connector on the other end to mate with system connectors 14P1 and 14J1 respectively. The other cable shall have a 577R590H59 connector on one end and a 579R256H64 connector on the other end to mate with system connectors 14P2 and 14J2 respectively.

(c) Indicator Control Cable Assembly #1 - This Cable Assembly shall permit normal operation within the system of the Indicator Control when it is removed from the aircraft. The Cable Assembly shall consist of one 579R564H02 connector on one end and one 579R256H02 connector on the other end to mate with system connectors 11P4 and 11J4 respectively. The Cable Assembly shall be 18 feet ± 6 inches in length.

(d) Indicator Control Cable Assembly #2 - This Cable Assembly shall permit normal operation within the system of the Indicator Control when it is removed from the aircraft. The Cable Assembly shall consist of three individual cables 18 feet ± 7 inches in length. The individual cables shall be formed into one main assembly leaving the individual cables free at both ends for a length of 18 ± 1 inches. One cable shall have a 577R590H64 connector at one end and one 579R256H59 connector at the other end to mate with system connectors 11P1 and 11J1 respectively. The second cable shall have a 577R590H65 connector at one end and one 579R256H60 connector at the other end to mate with system connectors 11P2 and 11J2 respectively. The third cable shall have a 577R590H36 connector at one end and a 579R256H31 connector at the other end to mate with system connectors 11P3 and 11J3 respectively.

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(e) Velocity Signal Computer Cable Assembly - This Cable Assembly shall permit normal operation within the system of the Velocity Signal Computer when it is removed from the aircraft to the Computer Rack (3.4.4). The Cable Assembly shall consist of three individual cables 18 feet ± 7 inches in length. The individual cables shall be formed into one main assembly leaving the individual cables free at both ends for a length of 18 ± 1 inches. One cable shall have a 577R590H65 connector at one end and one 579R256H60 connector at the other end to mate with system connectors 15P1 and 15J1 respectively. The second cable shall have a 577R590H66 connector at one end and one 579R256H61 connector at the other end to mate with the system connectors 15P2 and 15J2 respectively. The third cable shall have a 577R590H36 connector at one end and a 579R256H31 connector at the other end to mate with system connectors 15P3 and 15J3 respectively.

(f) Launch Signal Computer Cable Assembly - This Cable Assembly shall permit normal operation within the system of the Launch Signal Computer when it is removed from the aircraft to the Computer Rack (3.4.4). The Cable Assembly shall consist of three individual cables 18 feet ± 7 inches in length. The individual cables shall be formed into one main assembly leaving the individual cables free at both ends for a length of 18 ± 1 inches. One cable shall have a 577R590H64 connector at one end and one 579R256H59 connector at the other end to mate with system connectors 16P1 and 16J1 respectively. The second cable shall have a 577R590H67 connector at one end and one 579R256H62 connector at the other end to mate with system connectors 16P2 and 16J2 respectively. The third cable shall have a 577R590H37 connector at one end and a 579R256H32 connector at the other end to mate with system connectors 16P3 and 16J3 respectively.

(g) Pilot Command Signal Computer Cable Assembly - This Cable Assembly shall permit normal operation within the system of the Pilot Command Signal Computer when it is removed from the aircraft to the Computer Rack (3.4.4). The Cable Assembly shall consist of two individual cables 18 feet ± 7 inches in length. The individual cables shall be formed into one main assembly leaving the individual cables free at both ends for a length of 18 ± 1 inches. One cable shall have a 577R590H56 connector at one end and a

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579R256H51 connector at the other end to mate with system connectors 17P1 and 17J1 respectively. The other cable shall have a 577R590H55 connector on one end and a 579R256H50 connector on the other end to mate with system connectors 17P2 and 17J2 respectively.

(h) Antenna Control Cable Assembly - This Cable Assembly shall permit normal operation within the system of the Antenna Control when it is removed from the aircraft. The Cable Assembly shall consist of one 577R590H36 connector at one end and one 579R256H31 connector at the other end to mate with system connectors 13P1 and 13J1 respectively. The Cable Assembly shall be 4 feet ± 4 inches in length.

(i) BIT-FIM Cable Assembly - This Cable Assembly shall connect the Missile Control System Set, TS-2511/AWM-21 (FIM) (3.4.11) to the Selector-Test Programmer to permit operation of the FIM. The Cable Assembly shall have two connectors, 577R590H64 and 577R590H23, at one end and two 579R705H09 connectors at the other end to mate with system connectors 51P3, 51P2, 14P4 and 14P3 respectively. The Cable Assembly shall be 22 feet ± 14 inches in length.

3.4.5.6 Transit Case - The Transit Case shall be designed to hold all the Cable Assemblies (3.4.5.5). Compartments shall be provided by sheet aluminum partitions so that each Cable Assembly shall have a particular location within the Transit Case. The sides and partitions shall be lined with polyurethane foam per MIL-P-26514, Type I. The overall dimensions of the Transit Case shall not exceed 40 inches by 30 inches by 21 inches. The weight of the transit case shall not exceed 100 pounds.

3.4.6 Extension, Slide Rail Maintenance, MX-7345/AWM-22 (Extender Rail)

3.4.6.1 Function - The Extender Rail shall provide the slide rail necessary to extend the AN/APG-59 Radar Set Barrel Assembly beyond the aircraft for servicing and provide a means for attaching a cable to lift the AN/APG-59 Radar Set Barrel Assembly.

3.4.6.2 Form Factor - The Extender Rail shall be 51 ± 2 inches long. The outer end shall be a rail onto which the Radar Set Barrel Assembly can roll. The inner end of the Extender Rail shall be a plug 19 ± 2 inches long which inserts into the aircraft rail. The weight of the Extender Rail shall not exceed 12 pounds. The Extender Rail shall be of sufficient length to allow full withdrawal of the AN/APG-59 Radar Set Barrel Assembly when inserted into

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the aircraft rail. The insertion into the aircraft rail shall be a minimum of one half the length of the aircraft rail. The cross section of the Extender Rail shall be compatible with the aircraft rail. The outer end of the Extender Rail shall have a mechanical stop, which shall stop the Radar Set Barrel Assembly at the fully extended position. The stop shall have a resilient surface. A lock shall be provided on the Extender Rail to retain the Barrel Assembly in the fully extended position. A manual release mechanism shall be provided to permit the Barrel Assembly to be rolled back into the aircraft nose. Two screw pads shall be provided on the Extender Rail. Only one screw pad shall be used to hold the eye bolt used for lifting the AN/APG-59 Radar Set Barrel Assembly when it is attached to the Extender Rail. Both screw pads shall be used to hold the Coolant Filler and Bleeder (3.4.8). When the Extender Rail is fully seated into the aircraft rail, the Extender Rail shall release the Barrel Assembly lock to allow the Barrel Assembly to be withdrawn. Upon removal of the Extender Rail from the aircraft, the aircraft Barrel Assembly lock shall reseat, securing the Barrel Assembly. The Extender Rail lock shall retain the Extender Rail in position after it has been fully inserted into the aircraft rail. This lock shall be a manually operated release mechanism.

3.4.6.3 Service Conditions - The Extender Rail shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362, except that both shock and vibration shall be limited to 2 g with a Radar Set in place.

3.4.6.4 Safety - The following warning statement shall be painted on the Extender Bar:

WARNING

Latch Extender Rail firmly in place
before pulling out nose package.

Characters shall be 0.19 inches high and shall be applied by a black epoxy silk screen stenciling.

3.4.6.5 Associated Equipment

(a) The Extender Rail must engage with the aircraft rail and the AN/APG-59 Radar Set Rail.

(b) The Extender Rail must be capable of being stowed on the Maintenance Cart (3.4.3).

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3.4.7 Test Set, Antenna Alignment, TS-2516/AWM-22

3.4.7.1 Function - The Antenna Alignment Test Set shall accept radio frequency (rf) energy from the AN/APG-59 Radar Set and radiate this energy from a horn bearing a precise relation to a pair of optical targets to permit verification of the optical line of sight of the Radar Set Antenna, to check the zero elevation, zero azimuth position of the antenna gimbal to the antenna base and to energize the antenna with low level rf power.

3.4.7.2 Form Factor - The Antenna Alignment Test Set shall be housed in a Transit Case (3.4.7.4.6), with maximum dimensions of 36 inches by 24 inches by 15 inches and shall weigh 100 pounds maximum.

3.4.7.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.7.4 Contents of Unit - The test set shall consist of the following items:

(a) Tripod Assembly

(b) Cable Assemblies

(c) Telescope

(d) Antenna Base Mounting Fixture

3.4.7.4.1 Tripod Assembly - The Tripod Assembly shall consist of the Target and Horn Assembly, and the Tripod. The Tripod Assembly shall be capable of being folded into a package small enough to be easily handled by one man. Weight of the Assembly shall not exceed 25 pounds.

3.4.7.4.1.1 Target and Horn Assembly - Two optical targets shall be attached to the horn. Facing the horn, one optical target (Target A) shall be 2.060 inches up and 14.300 inches to the left of the horn, while the other (Target B) shall be 10.13 inches down and 2.31 inches to the left of the horn. The tolerances of the above dimensions shall be ±0.020 inches. The Target and Horn Assembly shall contain provisions for mounting the Telescope (3.4.7.4.4) so that the Assembly can be pointed directly at the AN/APG-59 Antenna. Vertical and horizontal lines shall be provided on the Assembly for roll adjustment. The Target and Horn Assembly shall be adjustable ±4 inches in a lateral direction, ±15 degrees in pitch and ±45 degrees in roll.

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3.4.7.4.1.1.1 Horn Specifications

Frequency Range:	X-Band
Gain:	15 dB minimum
Beamwidth:	30 degrees max
Connector:	Type N female
Polarization:	Vertical

3.4.7.4.1.1.2 Target Specifications

(a) Target A - The optical target shall be 4.00 ± 0.020 inches in diameter, with alternate quadrants of black and white. Superimposed on this pattern shall be a 2.00 ± 0.020 inches diameter circle with a 0.030 inch line width and a 0.480 ± 0.005 inches diameter spot, both of which are to be concentric with the bull's-eye to ±0.010 inches at their center.

(b) Target B - The optical target shall be 4.00 ± 0.020 inches in diameter with alternate quadrants of black and white. The black quadrants shall be upper left and lower right.

3.4.7.4.1.2 Tripod - The Tripod shall support the Target and Horn Assembly (3.4.7.4.1.1) and shall position the horn from between 3.5 to 6.5 ± 0.5 feet above the supporting surface.

3.4.7.4.2 Cable Assemblies - Two Cable Assemblies shall be provided, each of RG214/U per MIL-C-17, or equivalent. One Cable Assembly shall be 57 ± 1 feet in length with MS91236 connectors at both ends. The other Cable Assembly shall be 103 ± 1 feet with a MS91236 connector at one end and a MS91237 connector at the other end. The Cable Assemblies shall be marked at points of zero, 250 inches, 500 inches, 1000 inches, and 145 feet for locating the Tripod (3.4.7.4.1.1). The Cable markings shall start with the "zero" point marked on the 57-foot Cable at a nominal 8 feet from one end and continue onto the 103-foot Cable.

3.4.7.4.3 Antenna Base Mounting Fixture - This shall be a rigid Fixture that mounts to the lower foot of the AN/APG-59 Antenna base and holds the Telescope 3.4.7.4.4. The Fixture shall provide a Morse Taper Hole, the axis of which is parallel to both mounting planes within ±0.0002 radians to accept the Telescope. It shall be held on with captive screws.

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3.4.7.4.4 Telescope - A five power, elbow-type Telescope with an erect field of view, shall be supplied. The Telescope shall be provided with focusing adjustments for both range and ocular accommodation. It shall contain a reticle consisting of two concentric circles, 2 and 10 mils in diameter, crossed by two, centered perpendicular, hair lines. A tapered shaft, whose center line coincides with the optical line-of-sight, shall be fastened to the elbow junction of the Telescope. The shaft shall fit into a .50-caliber expander.

3.4.7.4.5 Coax to Waveguide Transition (Adapter) - A Coax to Waveguide Transition shall be supplied. The Adapter shall permit connection of the Cable Assembly (3.4.7.4.2) individually or in tandem to the AN/APG-59 Radar Set at the "f_o" output of the Radio Frequency Oscillator O-1330/APG-59. The Adapter shall be constructed from type RG67/U waveguide per MIL-W-85C. One end of the Adapter shall be terminated with a UG-136B/U Flange, modified per MIL-F-3992, and set at a 30° angle in the H plane. A stub 1.65 inches maximum overall length shall be attached at the other end of the Adapter. The end of the stub shall be equipped with a type N female connector, per MIL-C-71. The maximum length of the Adapter shall be 6 inches, with a minimum distance of 4.25 inches between the center line of the stub and the center line of the flange measured from its inner edge.

3.4.7.4.6 Transit Case - A Transit Case shall be provided to carry all the items of the Test Set. The maximum dimensions of the Transit Case shall be 36 inches by 24 inches by 15 inches. The weight of the Transit Case shall not exceed 50 pounds. Cushioning shall be provided in the Transit Case. The cushioning material shall be polyurethane foam charcoal gray color and 2 pound density per MIL-P-26514, Type I.

3.4.8 Filler and Bleeder, Coolant, MX-7346/AWM-22

3.4.8.1 Function - The Coolant Filler and Bleeder unit shall provide the means to replenish the coolant and bleed air from the AN/APG-59 Radar Set Liquid Cooling System. It shall also provide the means to initially charge the container.

3.4.8.2 Form Factor - The Coolant Filler and Bleeder shall be cylindrical in shape and shall have a minimum capacity of 400 cubic inches. The unit shall not exceed 15 inches diameter and 15 inches height. A frame made of welded wire shall support the liquid container in such a manner as to protect the container from mishandling. Provision shall be made on the frame for mounting the Coolant Filler and Bleeder to attachment points on the Slide Rail Maintenance Extension (3.4.6) during use. The frame shall provide a convenient handle for carrying. Brackets shall be provided for hose coilage when not in use. A filler cap shall be provided on top of the container and shall incorporate a standard Schrader aircheck valve fitting. A cylindrical strainer element of 250 micron nominal retention size, 2-1/4 inches diameter

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and 6 inches long, shall be fitted into the neck of the container. A thermometer, graduated from 0°F to +200°F, shall be provided to measure the coolant temperature with graduations accurate to ±5°F. The container shall have a minimum burst strength of 60 psig. A relief valve shall be provided which shall have a flow rate of 3 gpm at 50 psig and zero flow at 40 psig. The empty weight of the equipment, including Hose Assemblies (3.4.8.4), shall not exceed 25 pounds.

3.4.8.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-H-15362.

3.4.8.4 Hose Assemblies - The following hose assemblies shall be provided:

(a) Hose Assembly - The Hose Assembly shall be a rubber covered hose, 95 ± 6 inches in length, with quick disconnect, self-sealing fittings, a 9/16-18 NF-3 fitting at one end and a 577R715H02 coupling at the other end.

(b) Reversal Hose Assembly - This shall be a rubber covered hose, 5 ± 1 inches in length, and shall have a quick disconnect, self-sealing fitting, 577R715H02, at each end. A plate containing connection instructions shall be attached to the Hose Assembly.

3.4.9 Test Set, Vertical Gyro-Antenna Alignment, TS-2518/AWM-22

3.4.9.1 Function - The Vertical Gyro-Antenna Alignment Test Set shall provide the functions and facilities necessary to measure the alignment of the aircraft Vertical Reference System gyro with respect to the AN/APG-59 Antenna.

3.4.9.2 Form Factor - The Vertical Gyro-Antenna Alignment Test Set shall be a maximum of 12 inches by 14 inches by 12 inches. The weight of the equipment, including the transit case, shall not exceed 30 pounds.

3.4.9.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

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3.4.9.4 Contents of Unit - The Vertical Gyro-Antenna Alignment Test Set shall consist of the following units:

(a) Inclinometer

(b) Interconnecting Cable Assembly

(c) Feeler Gage

(d) Allen Wrench

(e) Transit Case

3.4.9.4.1 Inclinometer

3.4.9.4.1.1 Function - The Inclinometer shall sense the vertical and convert inputs from the aircraft Vertical Reference System gyro for measuring the alignment of the vertical gyro with respect to the AN/APG-59 Antenna.

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3.4.9.4.1.2 Form Factor - The Inclinometer shall have a Gravity Sensing Device (3.4.9.4.1.4) attached to the rotor of a Synchro Control Transformer (3.4.9.4.1.3) which shall be mounted to a casting forming the body of the Inclinometer. An arm extending from the body of the Inclinometer shall have a mounting pad which shall mount to the boresight pad of the AN/APG-59 Antenna in either of two planes, 90 degrees apart, to sense gravity in both the roll axis and the climb axis of the aircraft.

3.4.9.4.1.3 Synchro Control Transformer - The Synchro Control Transformer shall receive electrical signals from the vertical gyro and shall have a transformation ratio of 0.735 ±2%. The synchro shall be capable of being rotated in its mounting using a pinion wrench (BuOrd Dwg. 677036).

3.4.9.4.1.4 Gravity Sensing Device - Attached to the rotor of the Synchro Control Transformer (3.4.9.4.1.3), shall be a Gravity Sensing Device of sufficient weight to maintain a rotor constant with respect to earth when rotation, not exceeding ±18 degrees, is imparted to the stator. This device shall be damped with a damping factor of 0.8 ± 0.1. An external facility shall be provided to lock the rotor of the Synchro Control Transformer at 0 ± 0.05 degrees with respect to the mounting base when the stator is rotated for electrical zeroing and for protection during shipment.

3.4.9.4.1.5 Accuracy - The Inclinometer shall have an accuracy of ±6 minutes.

3.4.9.4.1.6 Input - The input connector shall be a connector MS3120E16-8P to receive S1, S2 and S3 signals from the aircraft Vertical Reference System gyro. A switch shall be provided to connect either the climb or roll signals to the Synchro Control Transformer (3.4.9.4.1.3).

3.4.9.4.1.7 Output - The rotor of the Synchro Control Transformer (3.4.9.4.1.3) shall be connected to connector J1.

3.4.9.4.2 Interconnecting Cable Assembly - The Interconnecting Cable Assembly shall be 33 ± 0.5 feet in length. One end of the Cable Assembly shall be terminated in a MS3126E16-8S connector to mate with the Inclinometer (3.4.9.4.1). Three connectors shall terminate the other end of the Cable Assembly as follows: a 577R590H53 connector shall mate with the aircraft Vertical Reference System gyro, one 577R568H01 connector shall mate with a meter, and a second 577R568H01 connector shall mate with the Antenna Control, C-7020/APG-59.

3.4.9.4.3 Feeler Gage - A 0.100 and 0.020 feeler gage shall be supplied to enable the gaps in the magnetic field to be set. The gap between the vane attached to the Gravity Sensing Device (3.4.9.4.1.4) and the magnet shall be adjustable by means of shims in the mounting of the magnet. The gap between the vane and the keeper shall be adjustable by movement of the

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Gravity Sensing Device on the axis of the rotor of the Synchro Control Transformer (3.4.9.4.1.3).

3.4.9.4.4 Allen Wrench - A standard 1/8 inch, Allen type wrench shall be provided for mounting the Inclinometer (3.4.9.4.1) on the boresight pad of the AN/APG-59 Antenna. A 1/16 inch, Allen type wrench shall be included for adjustment of the pendulum for proper damping vane clearance.

3.4.9.4.5 Transit Case - A transit case shall be provided and shall be a maximum of 12 inches by 14 inches by 12 inches. Cushioning material shall be provided in the case and shall be polyurethane foam in accordance with MIL-P-26514, Type I. The weight of the transit case shall not exceed 15 pounds.

3.4.10 Test Set, Reference Rate Gyro, TS-2512/AWM-21 - A Reference Rate Gyro Test Set shall be supplied.

3.4.11 Test Set, Missile Control System, TS-2511/AWM-21 - A Missile Control System Test Set shall be supplied.

3.4.12 Signal Generator, SG-696/AWM-21 - A Signal Generator shall be supplied.

3.4.13 Analyzer, Radio Frequency Noise, TS-2517/AWM-22 - A Radio Frequency Noise Analyzer shall be supplied.

3.4.14 Fixture, Antenna Transporting, MX-7339/AWM-21 - An Antenna Transporting Fixture shall be supplied.

3.4.15 RF Power Test Set, TS-1736/AWM-8 - A RF Power Test Set shall be supplied.

3.4.16 Compressor - Dehyrdator, Electronic Equipment 4D-755/AWM-21 - A Compressor - Dehydrator shall be supplied.

3.4.17 Counter, Electronic, Digital CP-870/AWM-21 - An Electronic shall be supplied in accordance with Electronic Counter, AN/USM-207, MIL-C-24165 less Converter, Frequency Plug-in.

3.4.18 Oscilloscope - An Oscilloscope shall be supplied.

3.4.19 Multimeter, AN/PSM-4 - A Multimeter in accordance with MIL-M-15134 is required for use with the AN/AWM-22. This Multimeter is general and standard equipment and is not supplied as part of the AN/AWM-22.

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3.4.20 Multimeter, TS-505/U - A Multimeter in accordance with MIL-M-55088 is required for use with the AN/AWM-22. This Multimeter is general and standard equipment and is not supplied as part of the AN/AWM-22.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(a) Preproduction Tests

(b) Acceptance Tests

(c) Life Tests

Note: Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(a) Contractor's Demonstration Tests

(b) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration test and when requested by the procuring activity the equipment shall be delivered to the Government for additional testing. This

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additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-5422 and interference tests in accordance with MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction models shall be by the Naval Air Systems Command upon satisfactory completion of all tests. All preproduction models, except those consumed in testing or those rendered irreparable for their intended use, shall be returned to the Contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction models shall not be considered as equipments under this contract; however, such models may be reworked and submitted for acceptance as production equipments. Incorporation of design changes, in delivered production equipments, found to be necessary as a result of the preproduction test program, shall be the subject of separate and independent negotiations on an individual basis.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

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(a) Examination of Product

(b) Operational Test

(c) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material, workmanship and interchangeability requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. Only those adjustments will be permitted which can be made by using such controls and adjustments as are accessible to the operator during the normal use of the equipment.

Temperature	Room Ambient
Humidity	Room Ambient
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum double amplitude of 0.036 inch.

The equipment shall be vibrated (without vibration isolators between the equipment and test fixtures) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be cycled periodically through its phases of operation. Should a failure occur, it should be repaired and tests started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record

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shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected for Sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2 additional
101 to 200	1 additional
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests after successful completion of the tests of 4.3.1.

(a) Complete operational tests at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Repeat the test of (a) above using the highest and lowest specified input voltages and frequencies.

(c) Operational test at environmental conditions when the results from the preproduction test indicate that these environmental conditions cause degradation(s) which result in marginal or unsatisfactory performance.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early production equipment.

(b) On an equipment midway of the production run.

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(c) On an early equipment after an extensive engineering change.

(d) When failure reports or other information indicate additional equipment should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control. The maximum number of special tests shall be defined by the procuring agency at the time of request for bids.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as are selected by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted:

(a) Temperature - Altitude Tests

(b) Vibration Tests (90 minutes cycling in each of 3 planes)

(c) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.1. The life tests sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

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Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1 additional
201 to 500	1 additional
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following conditions:

(a)	Temperature	Room Ambient
(b)	Altitude	Normal Ground
(c)	Humidity	Room Ambient

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.4 *Reconditioning of Life Test Samples* - An equipment which has been subject to the life test shall be reconditioned as follows:

(a) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedure - Preproduction tests, acceptance tests and life tests shall be performed in accordance with a test plan and procedures mutually agreeable to the procuring activity and the contractor. The test plan shall be prepared by the contractor. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, the procuring activity will furnish the procedures to be used for testing.

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4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished by the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability or interchangeability of maintenance parts of assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with MIL-T-23336.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, Number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

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(a) Contract - The contract shall have precedence over any specifications.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Notation of Revisions - A bar in the margin denotes a change made in that paragraph from the previous version of the document. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.4 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81516(AS) TEST SET, MISSILE CONTROL SYSTEM AN/AWM-22

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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